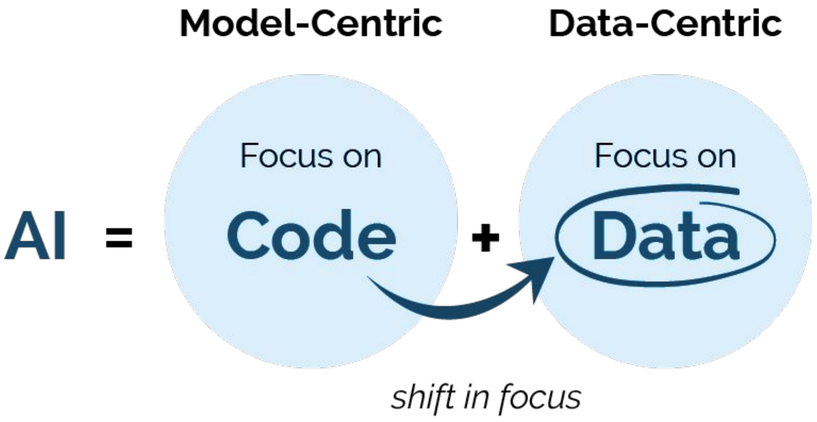
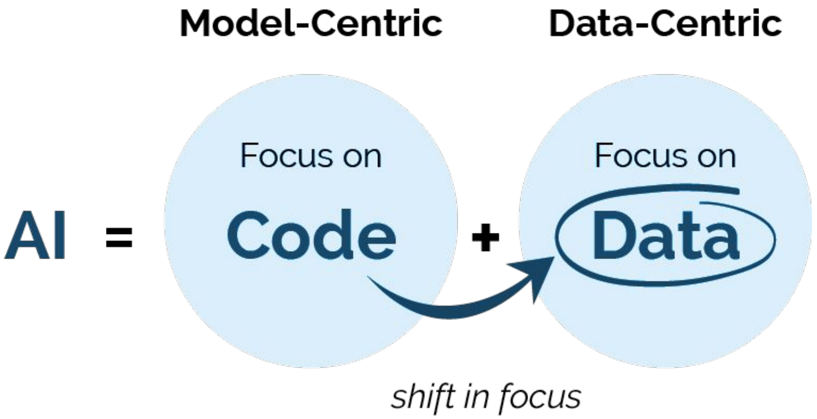


Data-centric AI: Perspectives and Challenges

이미지처리팀
류채은, 강인하, 안종식

Presenter contact info: superbunny38 [at] gmail [dot] com



Massachusetts Institute of Technology

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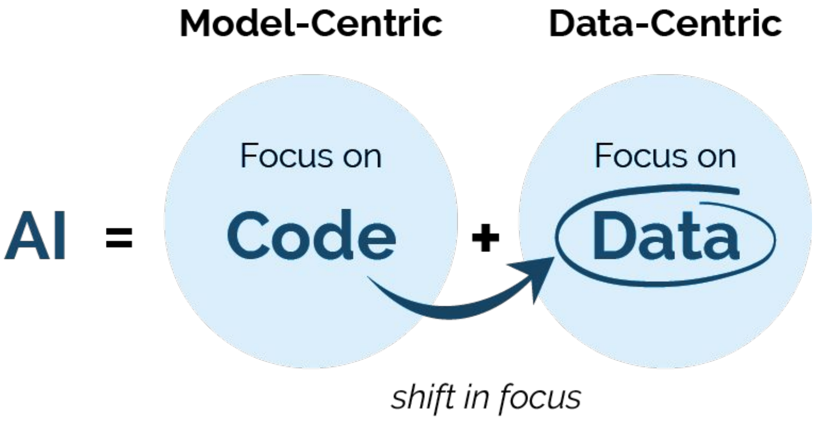

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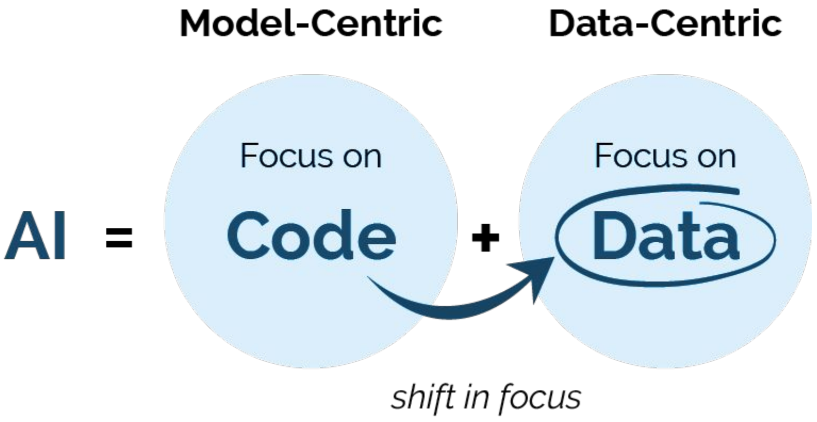

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Table of Contents

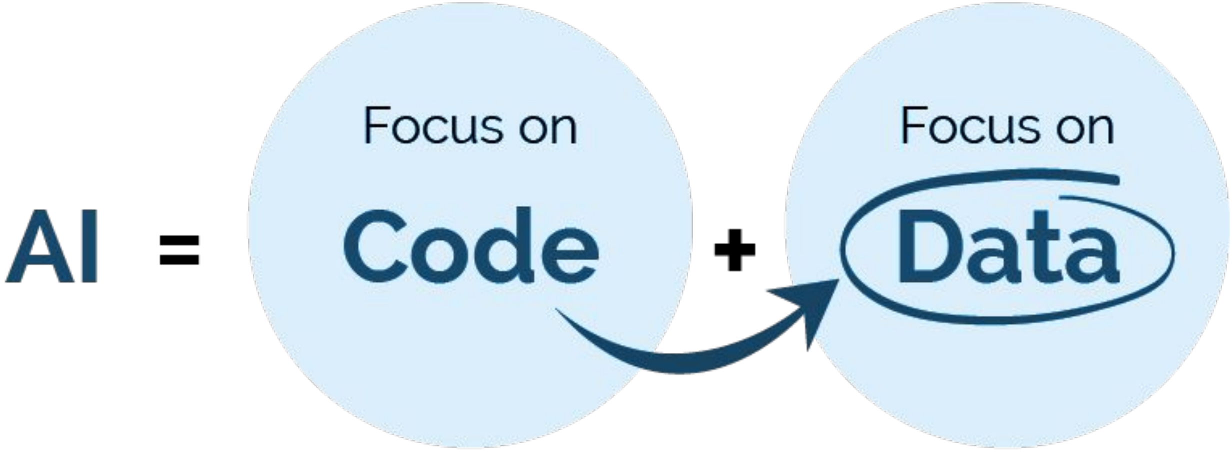
01

Motivation

Ref: https://www.youtube.com/watch?v=TU6u_T-s68Y

Motivation

Model-Centric **Data-Centric**



shift in focus

Ref: https://dcai.csail.mit.edu/lectures/data-centric-model-centric/

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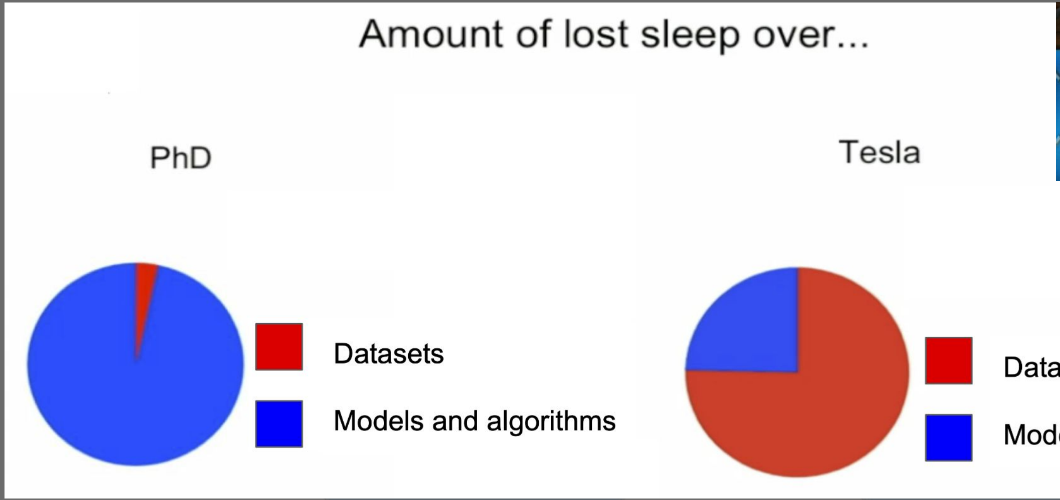

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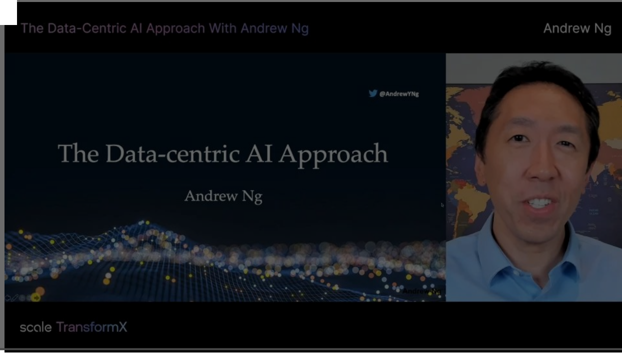

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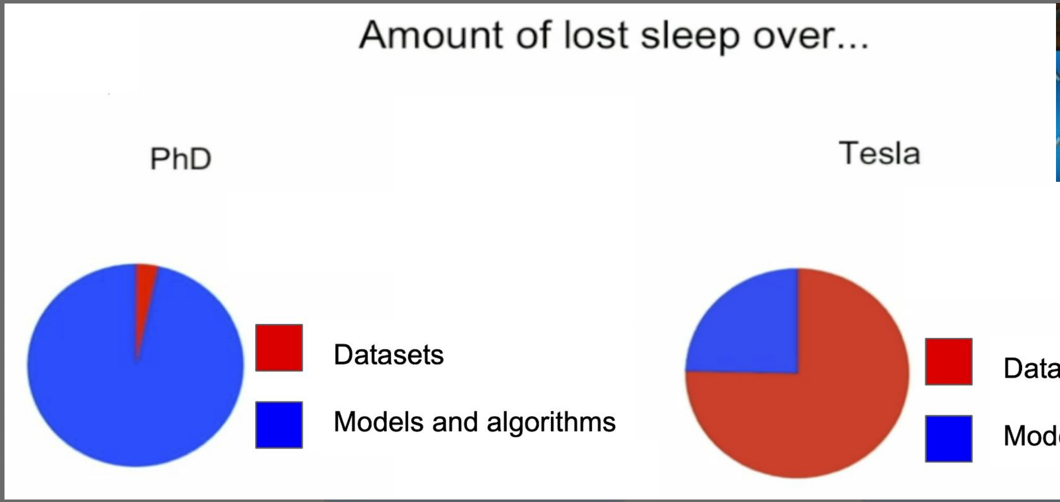

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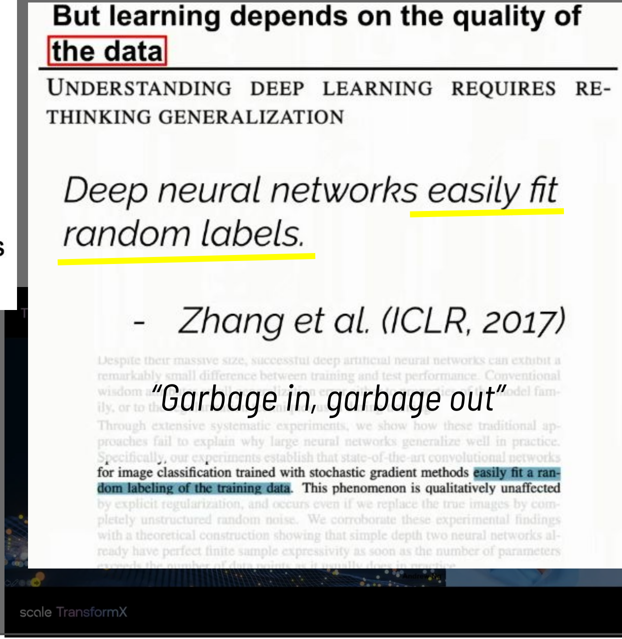
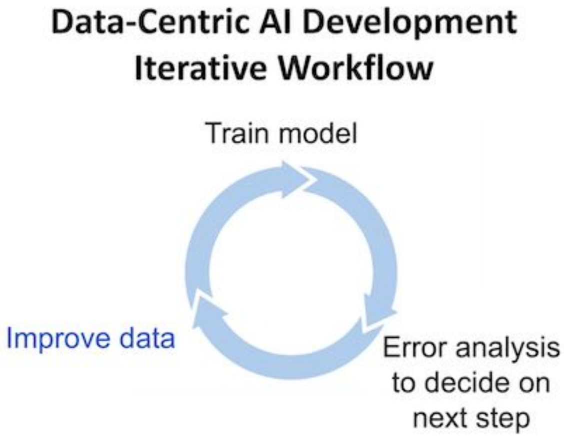


"Garbage in, garbage out"

Motivation


Data-Centric AI Development
Iterative Workflow
Train model
Improve data
Error analysis
to decide on
next step


The Data-Centric AI Approach With Andrew Ng
Andrew Ng
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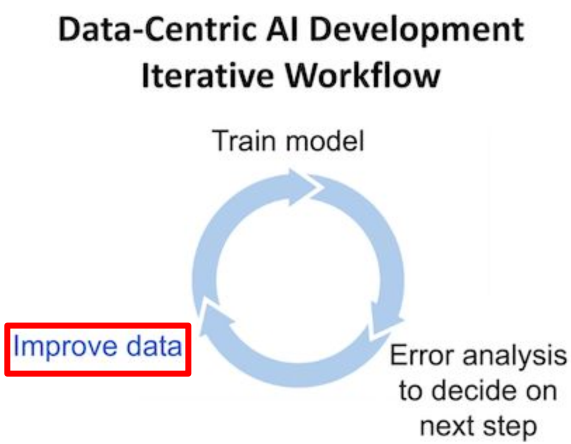
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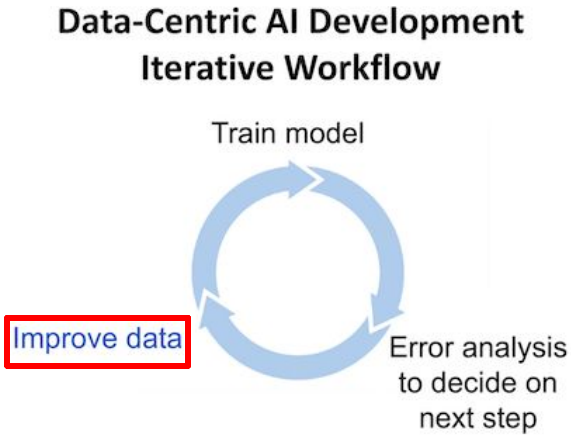

[Improve data (x,y) via these methods]

Improve y (=labels):

- Multiple labelers to measure consistency
- Improve label definitions and relabel more consistently

Improve x (ex. images):

- Toss out noisy examples or improve quality of input x
- Get more data through collection or data augmentation

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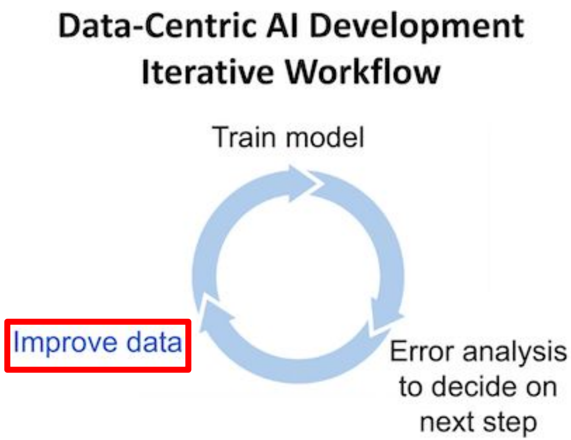

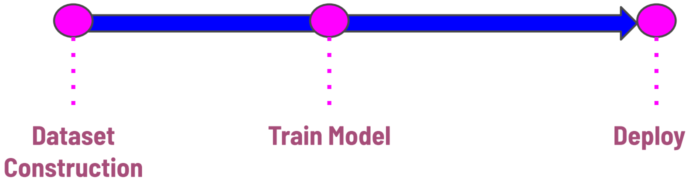

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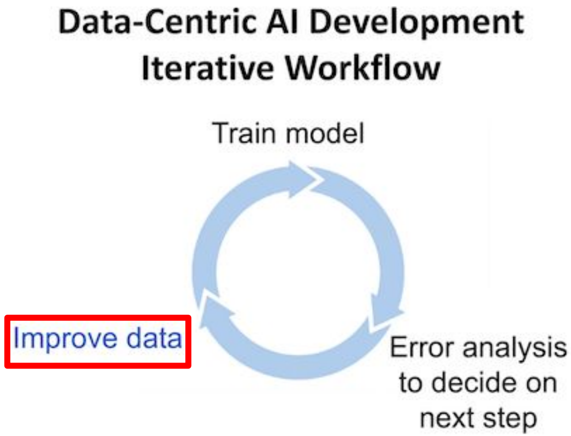

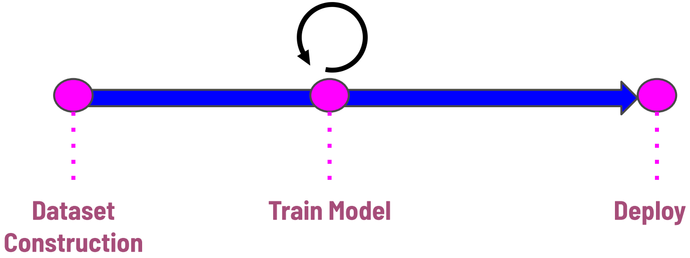

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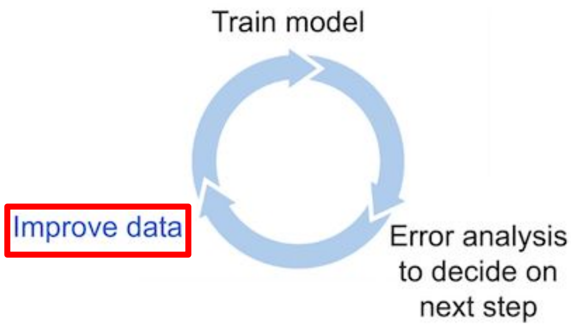
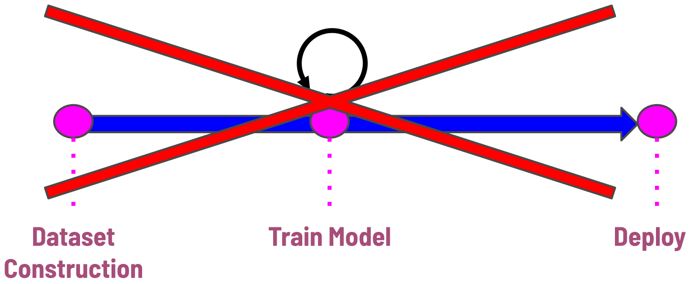
Motivation

Data-Centric AI Development
Iterative Workflow





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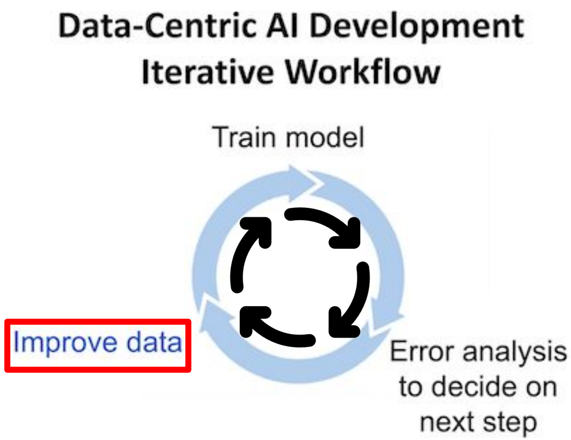

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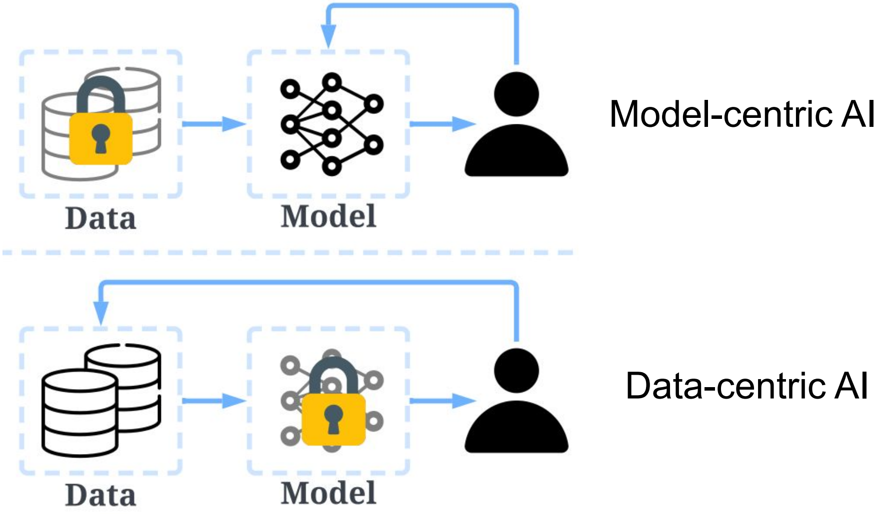
02

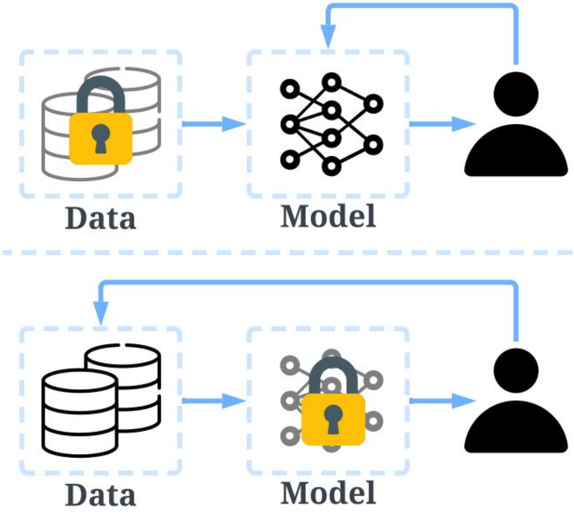
Introduction

Introduction



It trusts too much in data, which could be susceptible to flaws:
"Can the performance of model reflect the actual capability, or is it just overfitting the dataset?"

Introduction



Model-centric AI

Data-centric AI

Attention to **data** can help build **more powerful** AI to deal with **more complex real-world problems**

Introduction

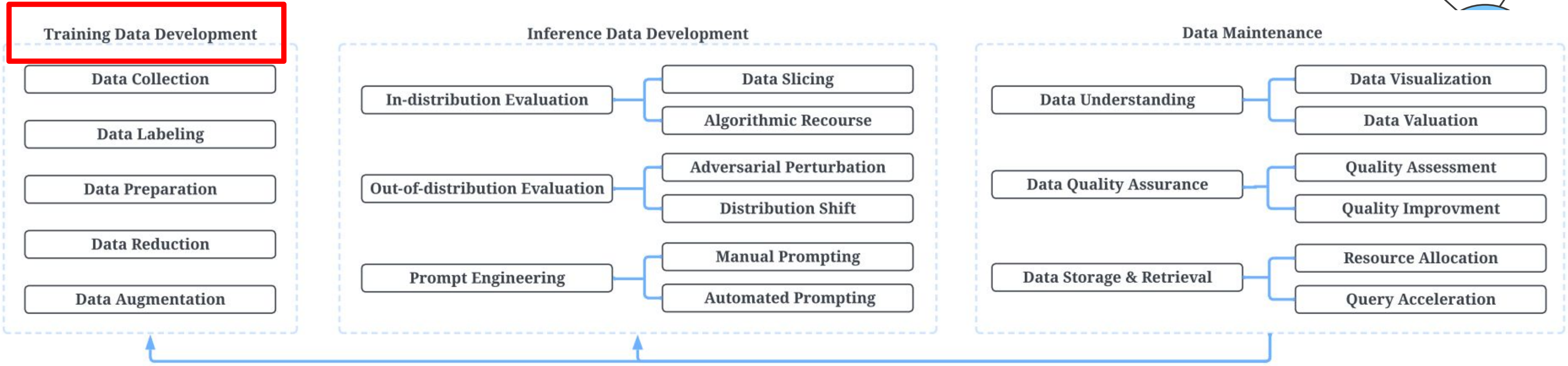


Figure 2: A big picture of DCAI missions and their representative tasks/subtasks.

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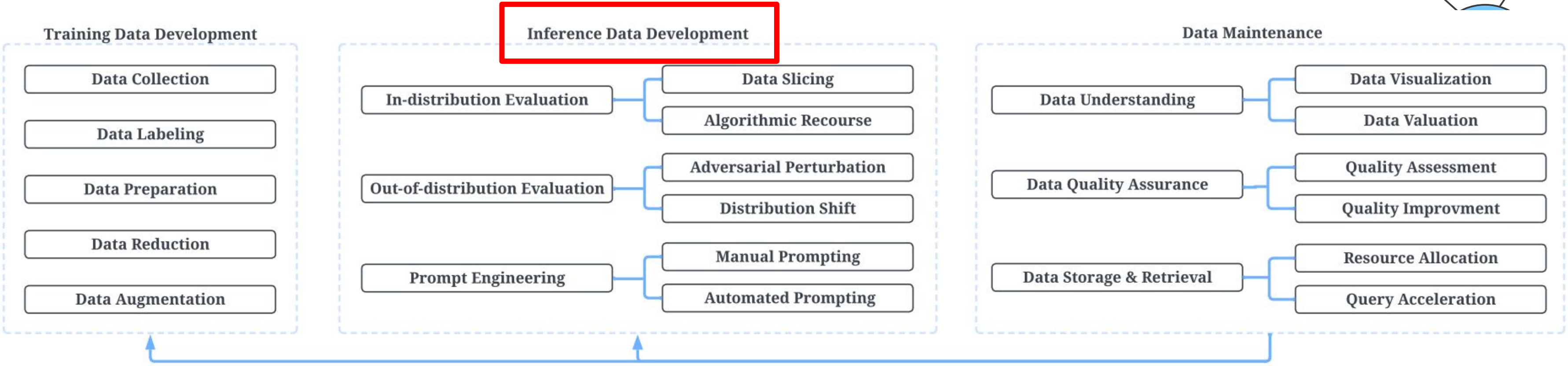

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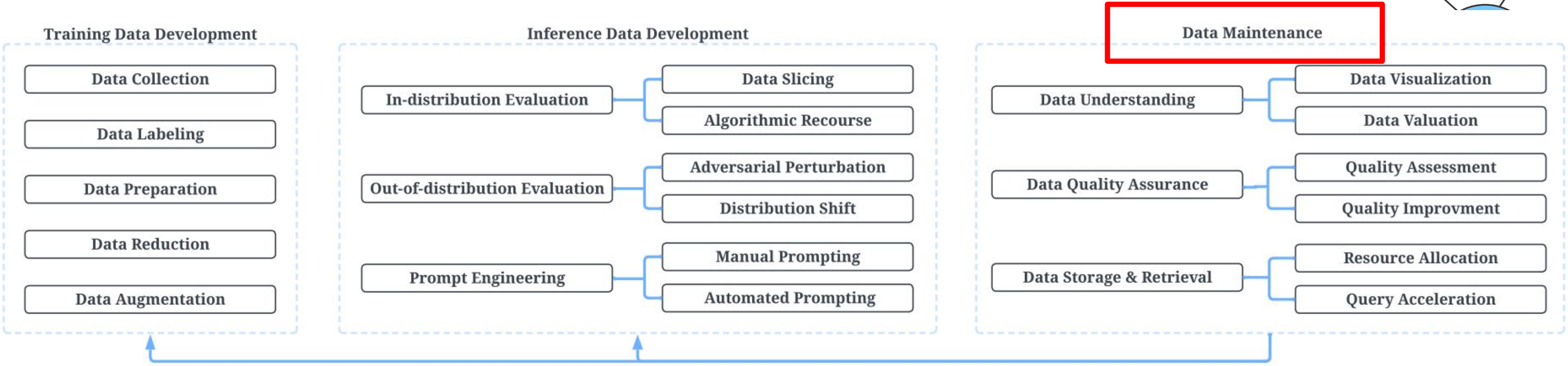

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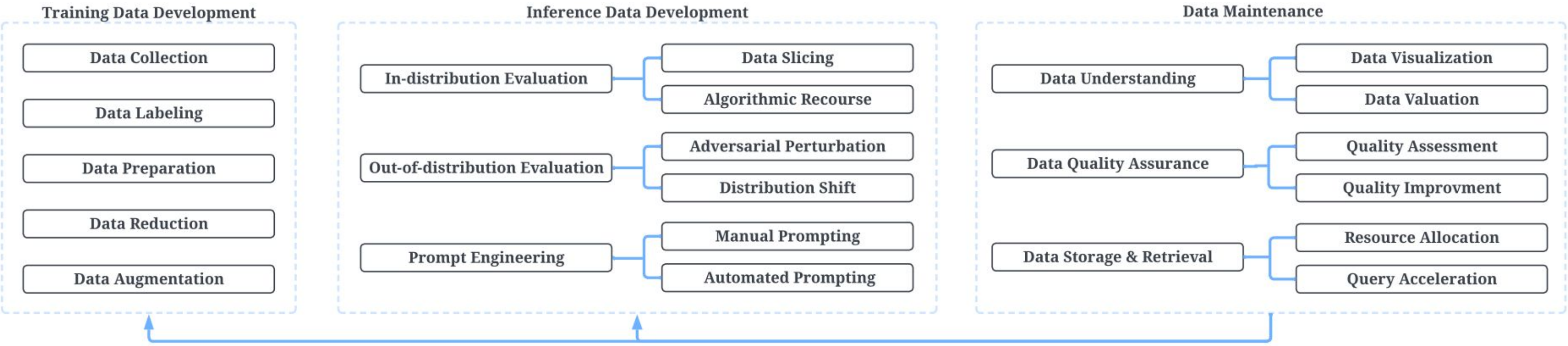


Figure 2: A big picture of DCAI missions and their representative tasks/subtasks.

Top-level summary of data-centric ai techniques!

03

Training Data Development

Training data development

5 Tasks in Training data construction:

1. Data collection
2. Data labeling
3. Data preparation
4. Data reduction
5. Data augmentation

Training data development

Data Collection

- **Dataset discovery:**
  Identify most relevant datasets from a raw format data lake!

- **Dataset Integration:**
  Combine multiple datasets into a larger dataset!

Training data development

Data Collection

- **Dataset discovery:**
  Identify most relevant datasets from a raw format data lake!

- **Dataset Integration:**
  Combine multiple datasets into a larger dataset!

Training data development

Data Collection

- **Dataset discovery**
- **Dataset Integration**

Data Labeling

- **Crowdsourcing**

“그냥 찍어야지…”

Ref: https://cleanlab.ai/blog/multiannotator/

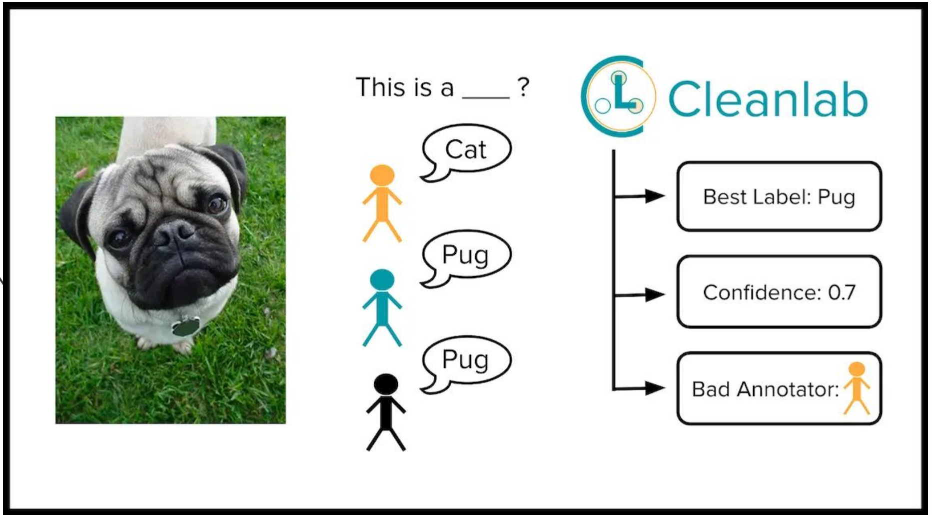
Training data development

Data Collection

- Dataset discovery
- Dataset Integration

Data Labeling

- Crowdsourcing



From CROWDLAB, you can get access to:

1. A **consensus label** for each example that aggregates the individual annotations (more accurately than majority-vote or algorithms used in crowdsourcing like Dawid-Skene).
2. A **quality score for each consensus label** which measures our confidence that this label is correct by considering the number of provided annotations and their agreement, prediction-confidence from a trained classifier, and trustworthiness of each annotator vs. the classifier.
3. A **quality score for each annotator** which estimates the overall correctness of their labels.

Ref: https://amitness.com/2020/07/semi-supervised-learning/

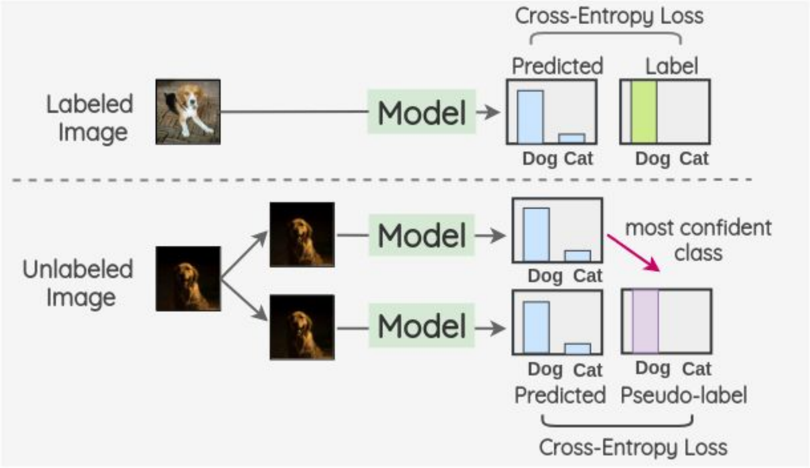
Training data development

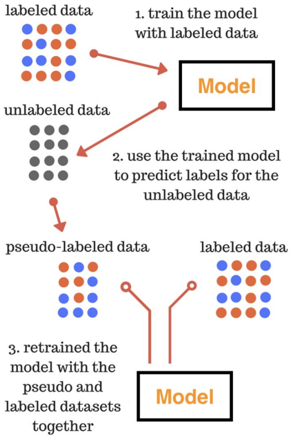
Data Collection

- Dataset discovery
- Dataset Integration

Data Labeling

- Crowdsourcing
- Semi-supervised Learning





Training data development

Data Collection

- Dataset discovery
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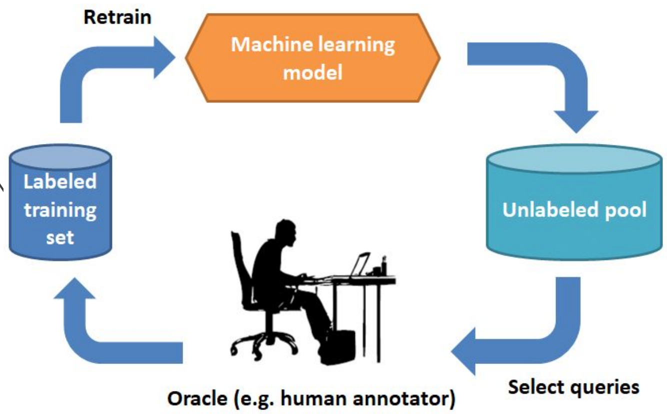
Data Labeling

- Crowdsourcing
- Semi-supervised Learning
- Active learning



Training data development

Data Collection

- Dataset discovery
- Dataset Integration

Data Labeling

- Crowdsourcing
- Semi-supervised Learning
- Active learning

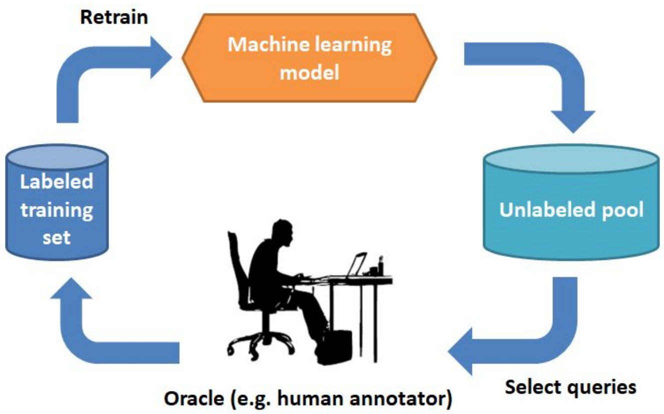
1-D: Passive vs. Active Learning

Active Learning (i=6)

Passive Learning (i=99)

active learning can give exponential speed-ups

Passive: err ~ n^{-1}
Active: err ~ 2^{-n}



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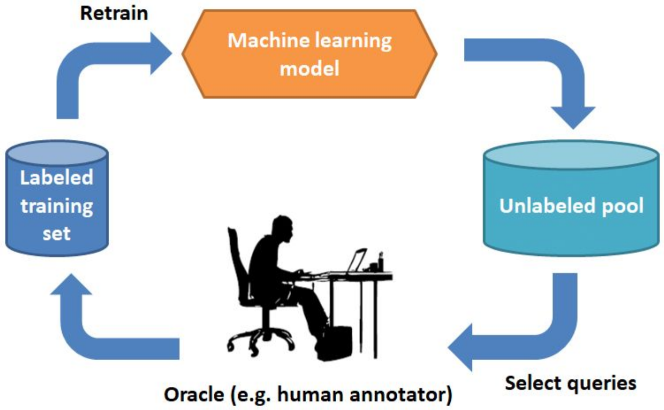

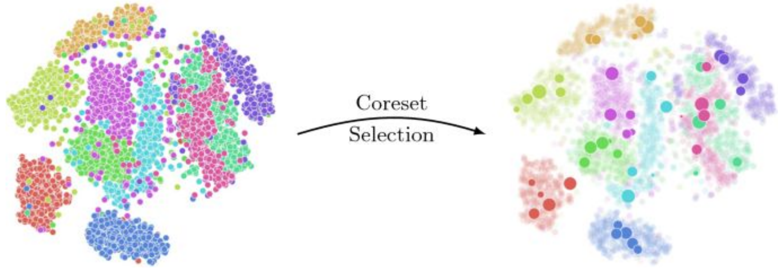

Training data development

Data Collection

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Ref:
- https://arxiv.org/abs/1911.00068
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Training data development

Data Collection
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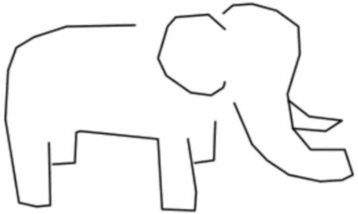
...

[Label Errors]



Dataset: **Google Quickdraw!**
Given Label: **Mosquito**

Dataset: **MNIST**
Given Label: **5**

"I've had this for over a year
and it works very well.
I am very happy with
this purchase."

Dataset: **Amazon Reviews**
Given Label: **1 star review**

...

Ref:
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Training data development

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...

[Confident Learning]

How does confident learning work?

Directly estimate the joint distribution of observed noisy labels and latent true labels.

$p(\tilde{y}|y^*)$, $p(y^*)$, $p(y^*|\tilde{y})$

$p(\tilde{y}, y^*)$	$y^*=dog$	$y^*=fox$	$y^*=cow$
$\tilde{y}=dog$	0.25	0.1	0.05
$\tilde{y}=fox$	0.14	0.15	0
$\tilde{y}=cow$	0.08	0.03	0.2

Off-diagonals tell you what fraction of your dataset is mislabeled.
Example -- "3% of your cow images are actually foxes"

Cleanlab

Open-Source

...

Baseline (remove prediction != label)	83.9	84.2
Confident learning methods	84.8	86.2
	86.7	86.9
	87.1	87.2
	87.1	87.2
INCV (Chen et al., 2019)	84.4	73.6
Mixup (Zhang et al., 2018)	76.1	59.8
SCE-loss (Wang et al., 2019)	76.3	58.3
MentorNet (Jiang et al., 2018)	64.4	61.5
Co-Teaching (Han et al., 2018)	62.9	58.1
S-Model (Goldberger et al., 2017)	58.6	57.5
Reed (Reed et al., 2015)	60.5	58.6
Baseline	60.2	57.3

Data-centric
Train with errors removed
"Change the dataset"

Same perf

Model-centric
Train with errors
"adjust the loss"

Perf drop-off

Training data development

Data Collection

- Dataset discovery
- Dataset Integration

Data Labeling

- Crowdsourcing
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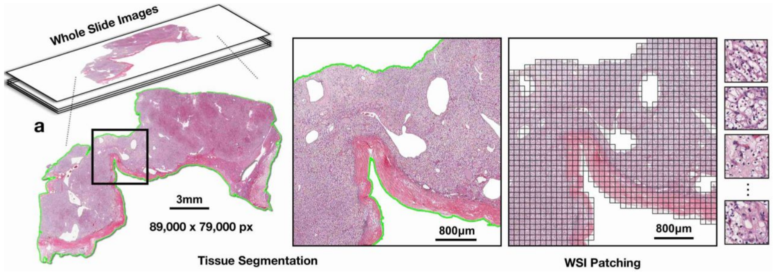
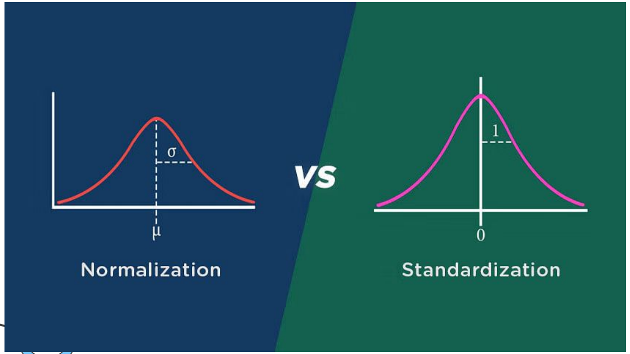
- Active learning

Data Preparation

- Data Cleaning
- Feature Extraction
- Data transformation

Divide Large Image into Small Patches





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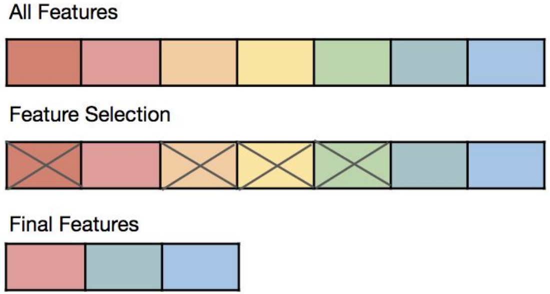
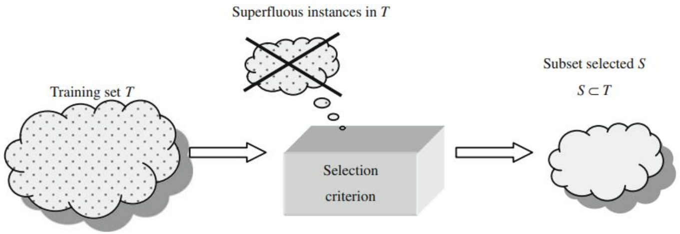
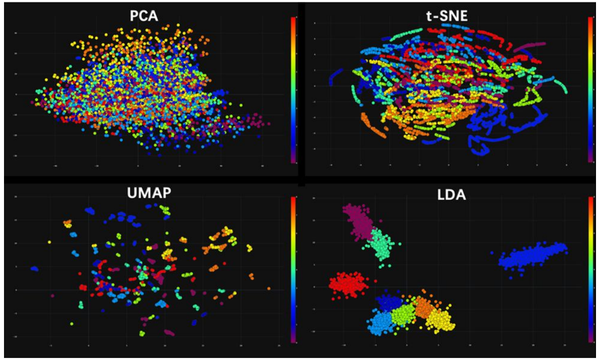
Data reduction

- Feature Selection
- Dimension Reduction
- Instance Selection







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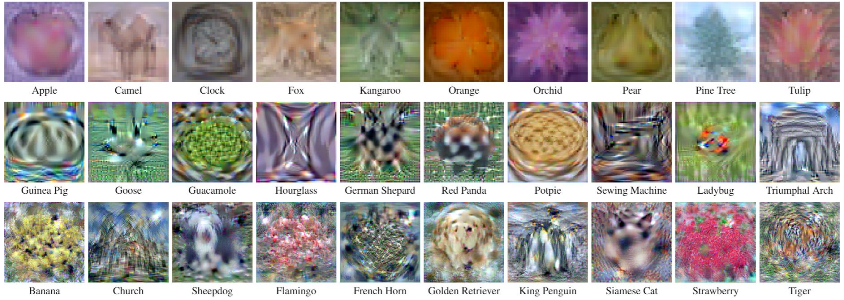
\+ Dataset Distillation



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Data augmentation

- Basic manipulation

Flip and rotation

Back-translation

Training data development

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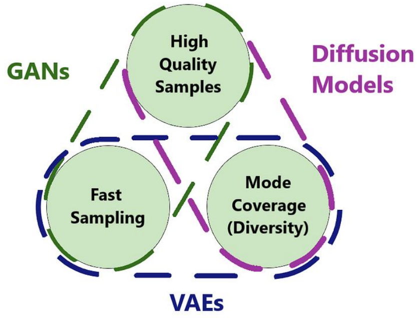
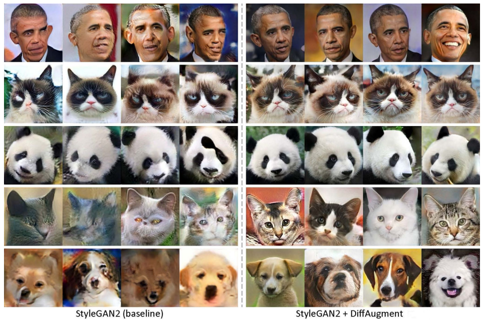
Data augmentation
- Basic manipulation
- Deep Learning Approaches





04

Inference Data Development

Inference data development

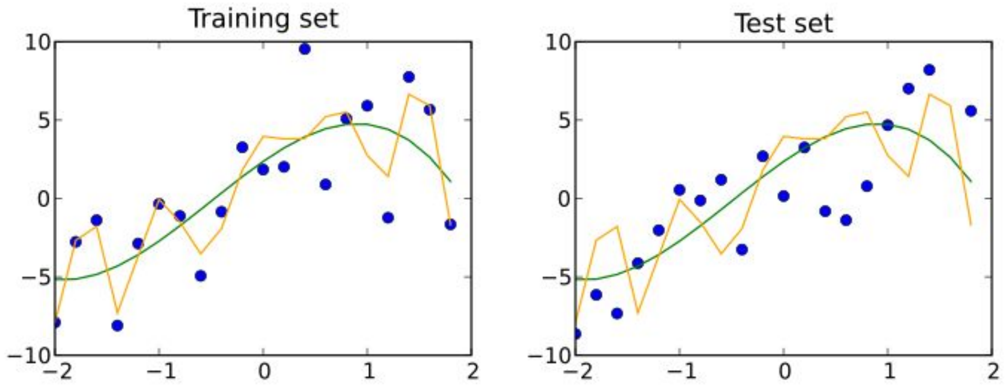
3 Related Tasks

1. In-distribution Evaluation
2. Out-of-distribution Evaluation
3. Prompt Engineering

Inference data development

In-distribution Evaluation Data

- **: Testing samples that follow the same distribution as the training data**



Inference data development

In-distribution Evaluation Data

- : Testing samples that follow the same distribution as the training data

Techniques for creating highly specific in-distribution set

1. **Data Slicing**
   data slice = a subset of the dataset that shares a common characteristic (=cohorts, subpopulation, or subgroup)

   example) data captured via: one sensor vs another, one location vs another

   -> Underperforming Subpopulations

Solution:

Trying more flexible models, oversampling, collect more data, and etc

Location A

Location B

Ref:
- https://github.com/HazyResearch/domino
- https://dcai.csail.mit.edu/lectures/data-centric-evaluation/data-centric-evaluation.pdf
-

Inference data development

In-distribution Evaluation Data

- : Testing samples that follow the same distribution as the training data

Techniques for creating highly specific in-distribution set

1. Data Slicing
   : a subset of the dataset that shares a common characteristic
   -> Underperforming Subpopulations

Slice Discovery Method

How to find subgroups in image dataset?

Ref:
- https://github.com/HazyResearch/domino
- https://dcai.csail.mit.edu/lectures/data-centric-evaluation/data-centric-evaluation.pdf
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Inference data development

In-distribution Evaluation Data

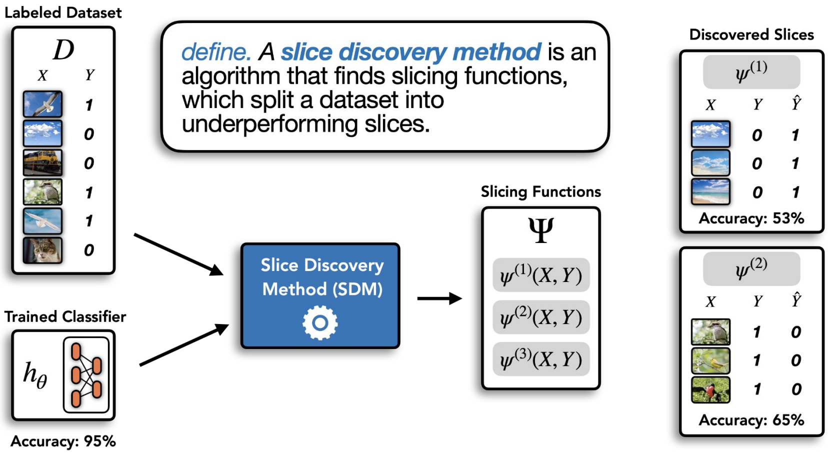
- : Testing samples that follow the same distribution as the training data

Techniques for creating highly specific in-distribution set

1. **Data Slicing**
   : a subset of the dataset that shares a common characteristic
   -> Underperforming Subpopulations

Slice Discovery Method

: Slice discovery is the task of mining unstructured input data (e.g. images, videos, audio) for semantically meaningful subgroups on which a model performs poorly.



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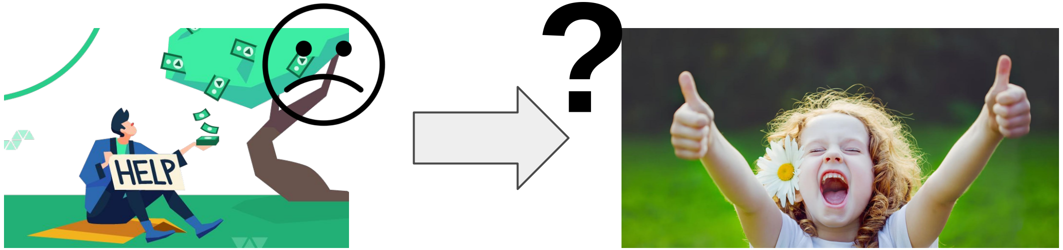
- https://dcai.csail.mit.edu/lectures/data-centric-evaluation/data-centric-evaluation.pdf
- https://slideslive.com/38942979/algorithmic-recourse-from-counterfactual-explanations-to-interventions
-

In-distribution Evaluation Data

2. **Algorithmic Resource**
   : A task to understand the decision boundaries of models.
   -> Find a hypothetical in-distribution set close to the model boundary but with different predictions.



[가정법 과거 had p.p]
만약 ~였다면 ~지
않았을텐데 (현재 반대)

Ex)
Consequence: " I applied for a loan"
Counterfactual explanation: "How the world would have had to be different for a desired outcome (=not applying for a loan) to occur
-> "If your salary had been $25,000 higher, you would not have applied for a loan"

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Inference data development

In-distribution Evaluation Data

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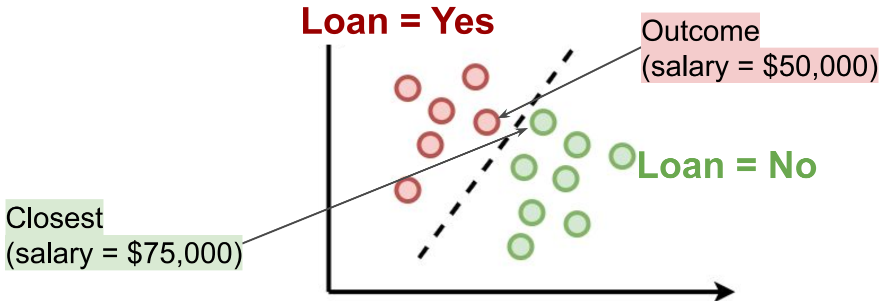

Ex)
Consequence: " I applied for a loan"
Counterfactual explanation: "How the world would have had to be different for a desired outcome (=not applying for a loan) to occur
-> "If your salary had been $25,000 higher, you would have been offered a loan"

Inference data development

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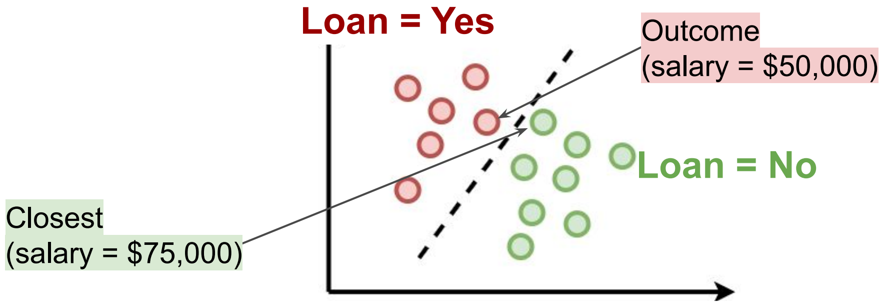


...

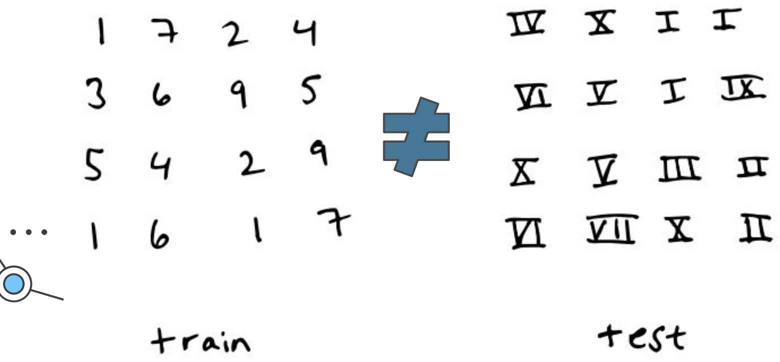
Helps explain predictions and prevent ethical issues and fatal errors!

Ref:
- https://dcai.csail.mit.edu/lectures/imbalance-outliers-shift/
-

Inference data development

Out-of-distribution Evaluation Data
: Testing samples follow a distribution that differs from the training data.

1. Adversarial perturbation: aims to design samples similar to the original ones but can mislead the model to test the robustness
2. Distribution shift: occurs when the joint distribution of inputs and outputs differs between training and test stages



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Inference data development

Out-of-distribution Evaluation Data
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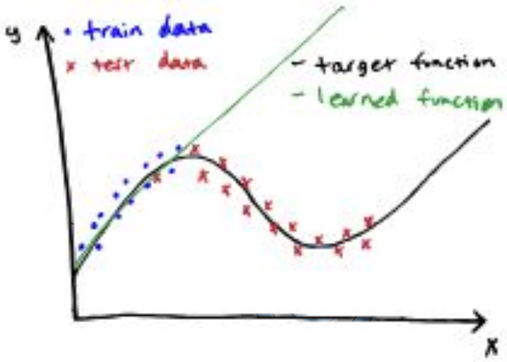
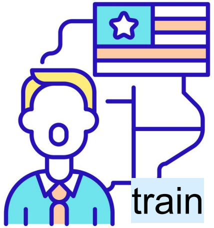
2. Distribution shift

[Covariate shift/data shift]




train

≠

test

The distribution of inputs changes between train and test, but the relationship between inputs and outputs does not change.

ex) Speech recognition model trained on native English speakers and then deployed for all English speakers

Ref:
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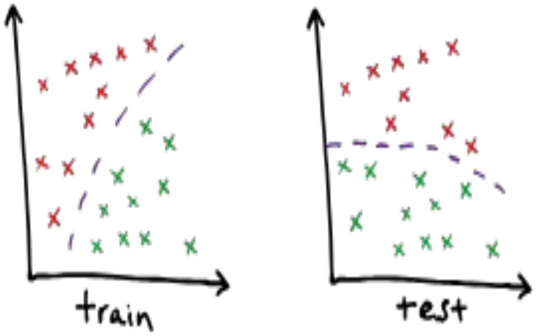
-

Inference data development

Out-of-distribution Evaluation Data
: Testing samples follow a distribution that differs from the training data.

2. Distribution shift

[Concept shift]



The input distribution does not change, but the relationship between inputs and outputs does.

ex) Making purchase recommendations based on web browsing behavior, trained on pre-pandemic data and deployed in March 2020. While web browsing behavior (x) did not change much (e.g., most individuals browsed the same websites, watched the same YouTube videos, etc.) before the pandemic vs during the pandemic, the relationship between browsing behavior and purchases did (e.g., someone who watched lots of travel videos on YouTube before the pandemic might buy plane or hotel tickets, while during the pandemic they might pay for nature documentary movies).

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Inference data development

Out-of-distribution Evaluation Data

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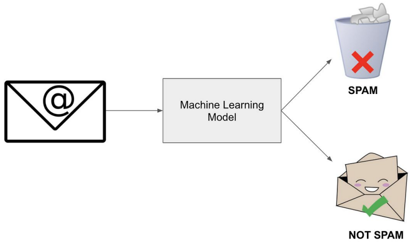
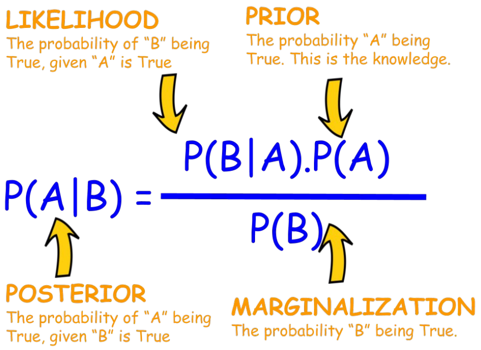


[Prior probability shift / label shift]



Occurs when p(y) changes between train and test data, but p(x|y) does not.

ex) If the model is trained on a balanced dataset of 50% spam and 50% non-spam emails, and then it's deployed in a real-world setting where 90% of emails are spam, that is an example of prior probability shift.

Inference data development

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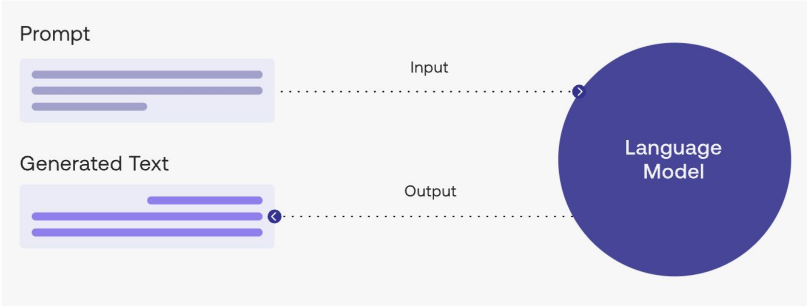
-

Prompt Engineering

: changing the input at test time to elicit certain results at output time.

ex)
- Prompt 1: Write a recommendation letter for a student
- Prompt 2: Write a recommendation letter for a student **for MIT**

...



05

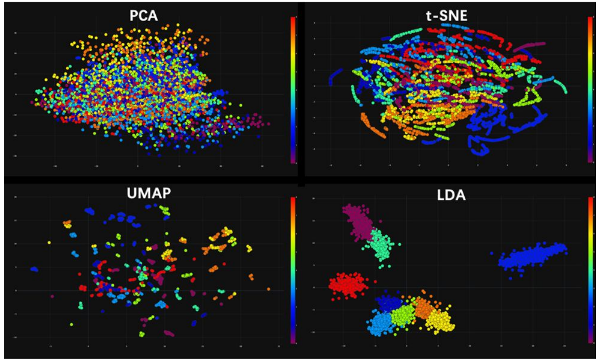
Data Maintenance

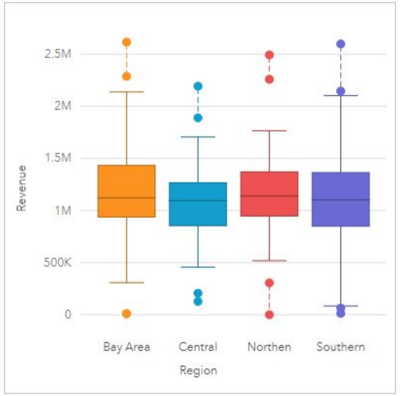
Data Maintenance

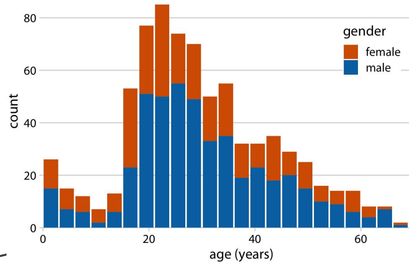
In production scenarios, data is not created once, but rather continuously updated.

Data understanding

- Data Visualization
- Data Valuation







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Data Maintenance

In production scenarios, data is not created once, but rather continuously updated.

Data understanding

- **Data Visualization**
- **Data Valuation**
  :helps understand what type of data is most valuable. It assigns credit to each data point regarding its contribution to the model performance.

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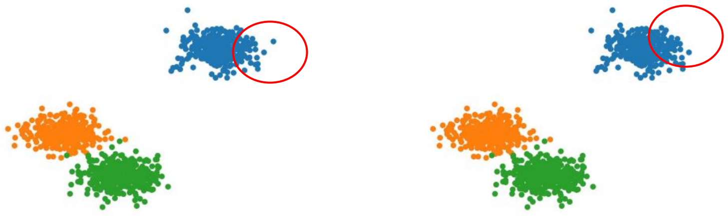
Leave-one-out Influence

How would model change if retrained after omitting datapoint (x, y) from dataset?



Trained Model has 98.5% validation accuracy

Trained Model has 98.3% validation accuracy

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Data Maintenance

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Data understanding

- Data Visualization
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Data quality assurance

- Quality assessment
- Quality improvement

Data Maintenance

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In production scenarios, data is not created once, but rather continuously updated.

Data understanding

- **Data Visualization**
- **Data Valuation**

Data quality assurance

- **Quality assessment**
- **Quality improvement**

Data acceleration

: aims to construct an efficient data infrastructure to facilitate the rapid acquisition of data

- **Resource allocation**
- **Query acceleration**

…

06
Open Research Challenges & Conclusion

Open Research Challenges & Conclusion

- **Inference Data & Data Maintenance**
  ： Rather than desperately optimizing performance metrics, they aim to provide a comprehensive understanding of the performance and continuous support of data.
- **Cross-task Techniques**
  ：Different DCAI tasks could have an interaction effect. It remains a challenge to systematically and simultaneously tackle data issues across multiple DCAI tasks.
- **Data-model Co-design**
  ：The optimal data strategies could differ when using different models and vice versa. With this in mind, our prediction is that future advancements will come from co-designing data pipelines and models and that the data-model co-evolvement will pave the way to more powerful AI systems.
- **Data Bias**
  ：Recently, AI systems in many highstake applications have been reported to show discrimination against certain groups of people, which raises serious fairness concerns. Mitigation of unfairness in data is necessary.
- **Benchmarks**
  ：It remains a challenge to construct a data benchmark to understand the overall data quality and comprehensively evaluate various DCAI techniques

- DCAI is an emerging research field that fundamentally shifts our focus from model to data.
- We provide a top-level discussion of its missions, aiming to help the community understand DCAI and push forward its progress.
- Many unsolved challenges remain to be addressed before we can fully achieve DCAI.

Thank you!
Any Questions?